

Heart System Documentation

Overview

The game now has proper UI heart sprites extracted from the official The Binding of Isaac: Rebirth sprite sheets. These hearts support multiple types for different health mechanics.

Heart Types

Red Hearts (Standard Health)

- **Full:** `heart_red_full` (index 0)
- **Half:** `heart_red_half` (index 1)
- **Empty:** `heart_red_empty` (index 2)

Red hearts are the standard health containers. Each full heart = 2 hit points.

Soul Hearts (Bonus Health)

- **Full:** `heart_soul_full` (index 3)
- **Half:** `heart_soul_half` (index 4)

Soul hearts float above red hearts and are consumed first when taking damage. They're typically blue/white in appearance.

Black Hearts (Special Health)

- **Full:** `heart_black_full` (index 5)

Black hearts provide 2 hit points and deal damage to all enemies in the room when depleted.

File Locations

Source Sprites (Original Size: ~16x15)

- `gfx_clean/ui/hearts/heart_red_full.png`
- `gfx_clean/ui/hearts/heart_red_half.png`
- `gfx_clean/ui/hearts/heart_red_empty.png`
- `gfx_clean/ui/hearts/heart_soul_full.png`
- `gfx_clean/ui/hearts/heart_soul_half.png`
- `gfx_clean/ui/hearts/heart_black_full.png`

Resized Sprites (2x Scale: 32x30)

Located in: `gfx_clean/resized/ui_hearts/`

These are scaled 2x using nearest-neighbor for pixel-perfect upscaling, making them more visible on the 3DS screen.

Atlas Integration

Atlas File: `gfx/ui_items.t3s`

The heart sprites are now included at the beginning of the UI items atlas. To rebuild the atlas after changes:

```
./rebuild_ui_atlas.sh
```

Or manually:

```
tex3ds -i gfx/ui_items.t3s -o romfs/ui_items.t3x
```

Header Definitions: `include/ui_items_atlas.h`

New heart indices:

```
#define ui_items_atlas_heart_red_full_idx 0
#define ui_items_atlas_heart_red_half_idx 1
#define ui_items_atlas_heart_red_empty_idx 2
#define ui_items_atlas_heart_soul_full_idx 3
#define ui_items_atlas_heart_soul_half_idx 4
#define ui_items_atlas_heart_black_full_idx 5
```

Legacy compatibility names (for existing code):

```
#define ui_items_atlas_ui_heart_full_idx 0 // = heart_red_full
#define ui_items_atlas_ui_heart_half_idx 1 // = heart_red_half
#define ui_items_atlas_ui_heart_empty_idx 2 // = heart_red_empty
```

Implementation Example

Drawing Hearts in Game Code

```
// Draw red hearts
for (int i = 0; i < game->player.max_health / 2; i++) {
    int heart_idx;
    int health_left = game->player.health - (i * 2);

    if (health_left >= 2) {
        heart_idx = ui_items_atlas_heart_red_full_idx;
    } else if (health_left == 1) {
        heart_idx = ui_items_atlas_heart_red_half_idx;
    } else {
        heart_idx = ui_items_atlas_heart_red_empty_idx;
    }

    C2D_DrawImageAt(
        C2D_SpriteSheetGetImage(ui_items_atlas, heart_idx),
        10 + (i * 34), // Spacing between hearts
        10,            // Y position at top
        0.5f           // Depth
    );
}

// Draw soul hearts
for (int i = 0; i < game->player.soul_hearts / 2; i++) {
    int heart_idx = (game->player.soul_hearts % 2 && i == game->player.soul_hearts /
2)
        ? ui_items_atlas_heart_soul_half_idx
        : ui_items_atlas_heart_soul_full_idx;

    C2D_DrawImageAt(
        C2D_SpriteSheetGetImage(ui_items_atlas, heart_idx),
        10 + ((game->player.max_health / 2 + i) * 34),
        10,
        0.5f
    );
}
```

Future Enhancements

Additional Heart Types (Not Yet Implemented)

The original game includes:

- **Eternal Hearts:** White hearts that become permanent after clearing a floor
- **Bone Hearts:** Red outlined hearts that slowly recharge when empty
- **Rotten Hearts:** Green hearts that occupy a red heart container
- **Golden Hearts:** Temporary hearts that prevent damage

These can be added by:

1. Extracting sprites from the official sprite sheets
2. Adding them to `gfx_clean/resized/ui_hearts/`
3. Updating `gfx/ui_items.t3s`
4. Regenerating the atlas
5. Adding defines to `include/ui_items_atlas.h`

Technical Notes

- **Sprite Size:** Original hearts are ~16x15 pixels, scaled 2x to 32x30 for 3DS
- **Format:** RGBA8888 for proper transparency
- **Spacing:** Hearts should be spaced ~34 pixels apart horizontally (32px width + 2px gap)
- **Z-Depth:** Draw at 0.5f depth to appear above gameplay but below text
- **Performance:** Hearts are pre-rendered in the atlas, so drawing them is very efficient

Credits

Heart sprites extracted from The Binding of Isaac: Rebirth official sprite sheets.

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